Case Study 2 Report

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Stat 311

10/31/2025

1. Introduction

Research Question

This case study investigates a fundamental question in real estate valuation: **How do appraised property values (land and improvements) and neighborhood location collectively influence residential sale prices, and does this relationship remain consistent across different neighborhoods?**

More specifically, we aim to determine:

- Whether appraised land and improvement values are reliable predictors of market sale prices?
- How much does neighborhood characteristics contribute to property valuation beyond physical assets?
- Whether appraisers apply consistent valuation criteria across sociologically and economically diverse neighborhoods?

Data Collection Purpose and Relevance

The data was supplied by the Hillsborough County Property Appraiser's office (Florida) to address critical questions about property assessment, accuracy and equity. As the textbook highlights, if the relationship between appraised values and sale prices differs significantly by neighborhood, this suggests inconsistent appraisal practices, with profound implications for tax fairness, investment strategy, and market efficiency.

The data represents residential sales from January 2017 through January 2018, extending previous research from 4 to 12 neighborhoods for a comprehensive analysis during a stable real estate market.

Relevance to Real Estate Professionals:

This analysis is crucial for multiple stakeholders:

- Property Appraisers: Validate assessment methodologies across neighborhoods.
- Real Estate Investors: Identify neighborhoods with potential undervaluation.
- Homebuyers: Understand how property characteristics translate to market value.
- Tax Assessors: Ensure equitable property taxation across regions.

2. Data and Methods

Study Design and Data Description

The dataset contains 1,264 residential property transactions across 12 neighborhoods in Tampa, Florida. Key variables include:

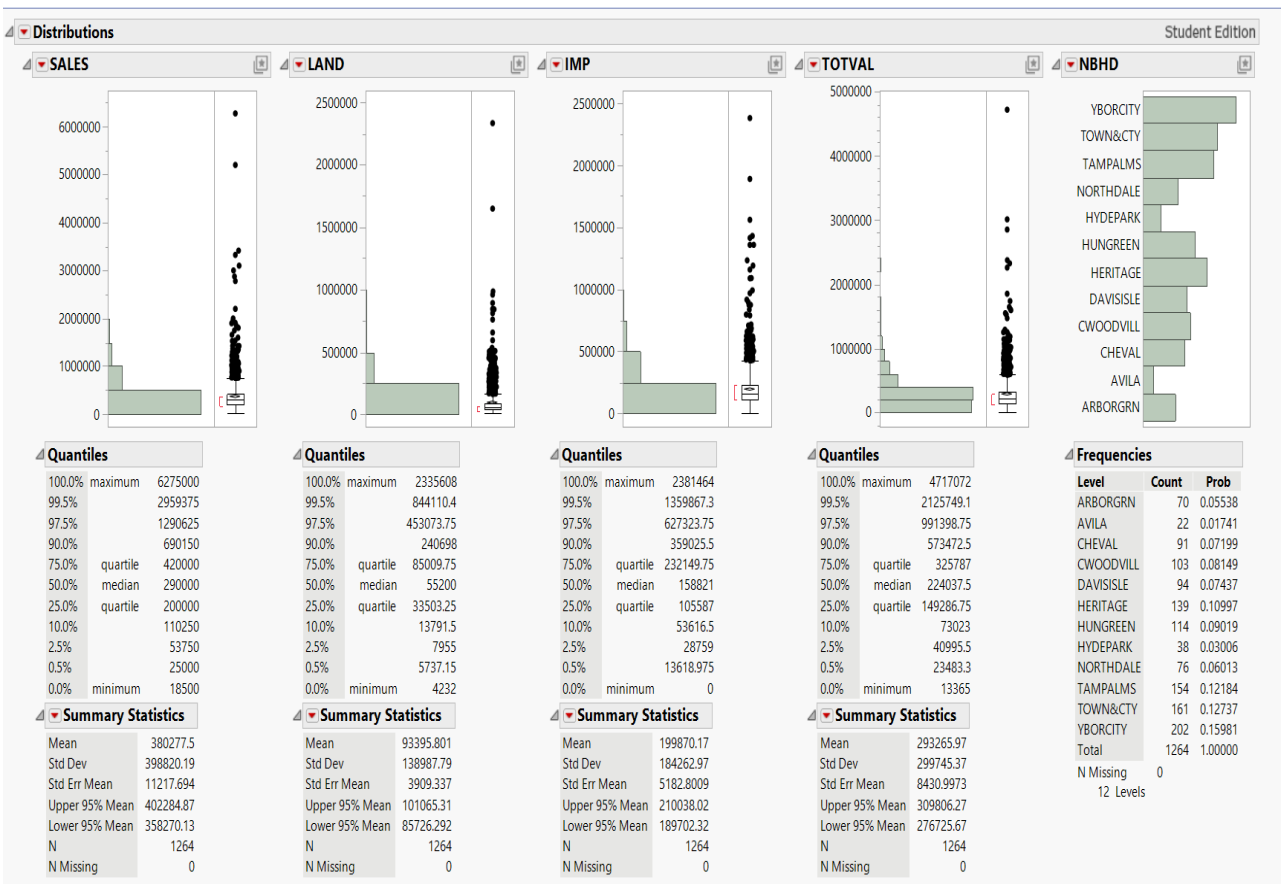
Dependent Variable: SALES (Sale price in dollars)

Independent Variables:

- **LAND**: Appraised land value (dollars)
- **IMP**: Appraised improvement value (dollars)
- **NBHD**: Neighborhood (12 categories: ARBORGRN, AVILA, CHEVAL, CWOODVILL, DAVISISLE, HERITAGE, HUNGREEN, HYDEPARK, NORTHDALE, TAMPALMS, TOWN&CTY, YBORCITY)

Data Cleaning: The dataset was imported into JMP with NBHD coded as a nominal variable. No missing values were detected in key analysis variables.

Figure 1: **Distribution of Sale Prices by Neighborhood**



This figure shows the distribution of sale prices across neighborhoods. It highlights the variation in market values due to location, supporting the inclusion of NBHD as a categorical variable in the model.

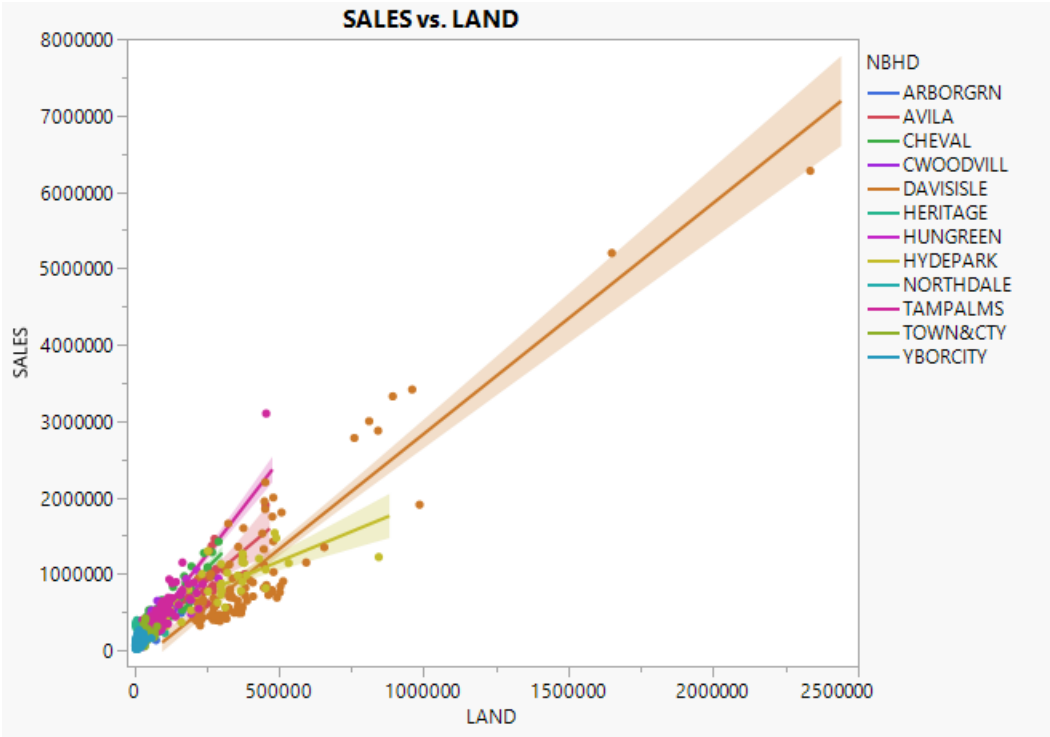
Summary Statistics LAND,IMP, AND SALES by Neighborhood

Tabulate Student Edition

	LAND				IMP				SALES				TOTVAL			
NBHD	Mean	Std Dev	Min	Max	Mean	Std Dev	Min	Max	Mean	Std Dev	Min	Max	Mean	Std Dev	Min	Max
ARBORGRN	72294.014286	22543.757551	37895	162162	209292.95714	44812.561227	131492	291754	357910.41429	85856.275357	135000	515000	281586.97143	62894.629492	173529	430648
AVILA	271918.54545	66530.42088	159782	453600	501509.04545	216283.38842	268597	970359	973436.36364	318515.30358	615000	1900000	773427.59091	250340.62175	469558	1252504
CHEVAL	99642.142857	56362.794475	37646	291519	301837.07692	173169.18213	140847	1235275	497476.92308	240926.13856	202000	1425000	401479.21978	221710.48335	182823	1526794
CWOODVILL	68292.15534	22450.324675	28875	161523	224424.83495	72543.834115	113448	476562	372263.1068	101701.53853	47800	750000	292716.99029	88724.891541	142323	638085
DAVISISLE	417504.89362	288520.22076	203136	2335608	392809.59574	455359.46848	24795	2381464	1074008.5106	959320.05957	325000	6275000	810314.48936	693974.41059	251253	4717072
HERITAGE	51431.417266	20015.390675	6250	114569	158197.05755	40431.477696	80312	248991	276006.68345	58907.33096	158000	470000	209628.47482	48939.201053	108716	363560
HUNGREEN	65612.008772	36935.729066	30087	290716	200977.32456	93797.813834	97689	671956	345735.96491	140790.12255	175000	1050000	266589.33333	126849.40759	131161	962285
HYDEPARK	311737.34211	145965.46745	110124	844906	335394.21053	160290.51221	106405	631635	871105.26316	311113.51621	319000	1537500	647131.55263	256973.81786	237023	1150768
NORTHDALE	50390.381579	11705.025544	32616	91919	167512.63158	38691.590204	107516	285252	276776.31579	37802.313398	155000	357500	217903.01316	41122.674632	143254	332301
TAMPALMS	83330.363636	49498.091735	28671	455872	260913.24026	138228.83909	105573	1092222	428696.1039	276391.43595	190000	3100000	344243.6039	182819.57111	153203	1548094
TOWN&CTY	34631.956522	8869.94266	26010	78951	109629.37888	38697.926267	48813	353603	185047.82609	49893.056991	54200	420000	144261.3354	42822.025804	74823	391763
YBORCITY	14601.346535	6998.7364609	4232	58721	55630.881188	29460.992949	0	155486	108432.67327	53649.962066	18500	264000	70232.227723	31731.496563	13365	169409

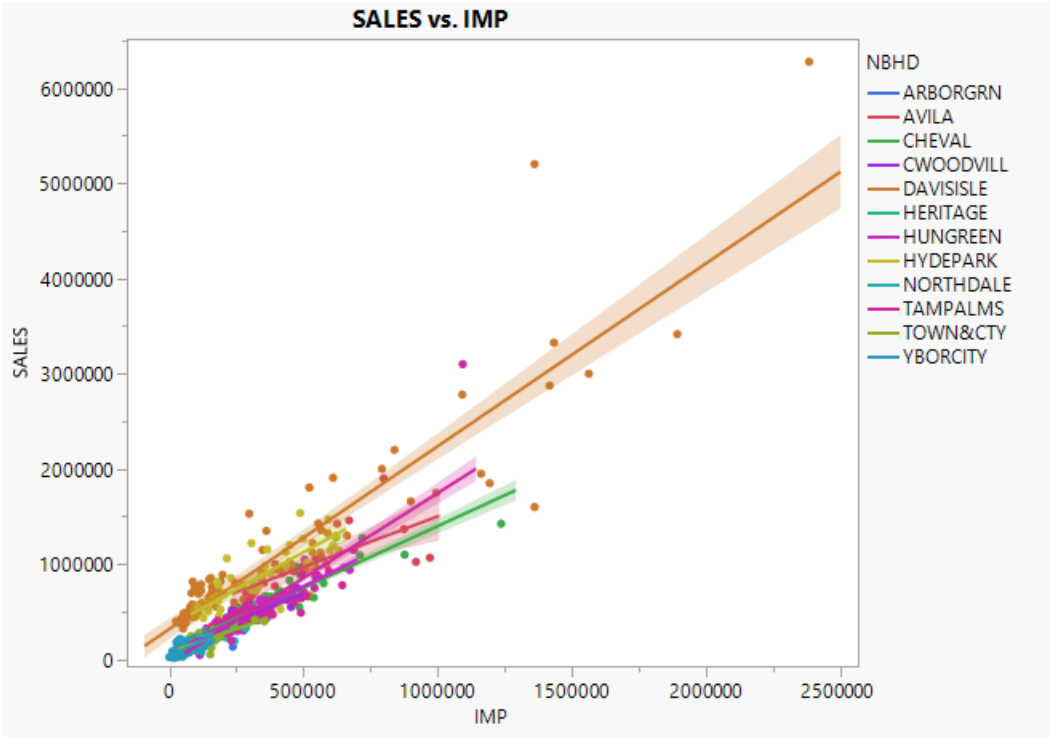
These statistics provide a numerical summary of sale prices by neighborhood, showing differences in central tendency and spread. This supports the hypothesis that location affects price.

Figure 2: SALES vs LAND colored by NBHD



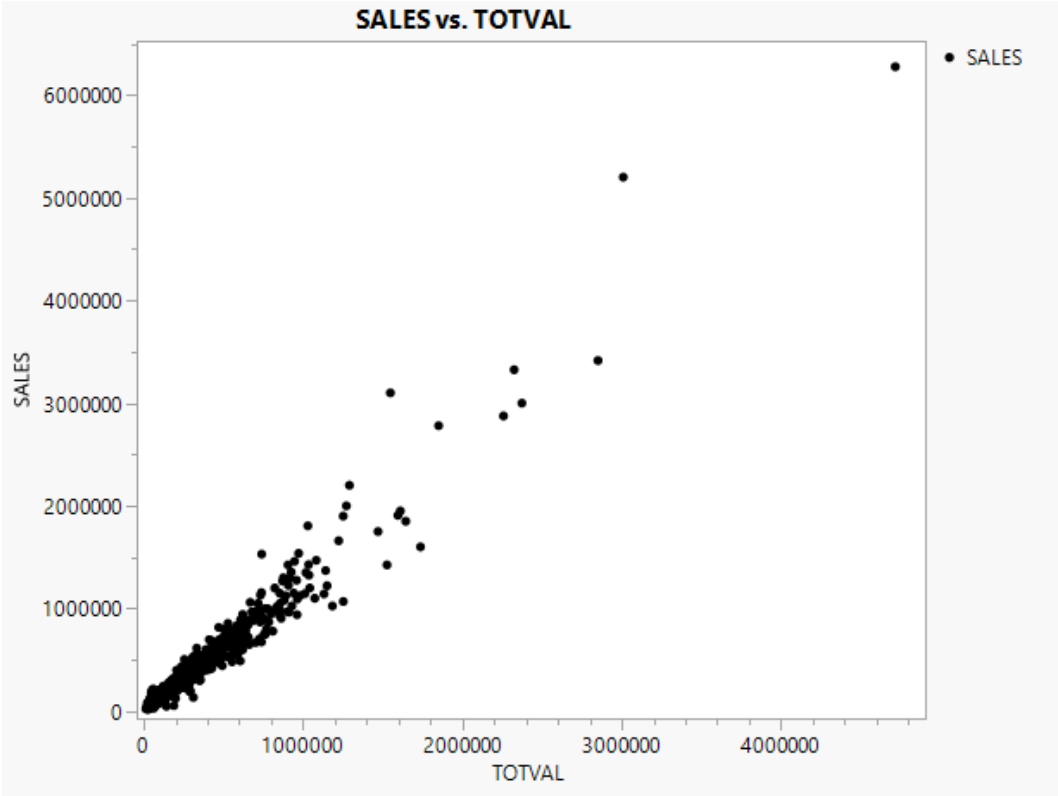
The scatterplot of SALES vs LAND colored by NBHD shows a generally positive relationship between land value and sale price. The color coding reveals that this relationship is land effects vary by neighborhoods, justifying the use of LAND as a predictor.

Figure 3: SALES vs IMP colored by NBHD



The scatterplot of SALES vs IMP colored by NBHD shows a positive relationship between improvement value and sale price. The improvement effects vary by neighborhood.

Figure 4: SALES vs TOTVAL



The scatterplot of SALES vs TOTVAL (LAND + IMP) shows a strong linear relationship, indicating that the combined appraised value is a good predictor of sale price.

Statistical Methods: Model Comparison Approach

A hierarchical model-building approach was employed to determine the most parsimonious yet accurate model. This involved testing increasingly complex models and using formal statistical tests to determine when additional complexity was justified.

Research Questions:

1. Do appraised land and improvement values predict sale prices?

2. Do neighborhood differences affect sale prices?

3. Does the relationship between appraised values and sale prices vary by neighborhood?

4. What is the most parsimonious model that adequately explains sale price variation?

Regression Models (all variables in thousands of dollars):

Model 1: First-order main effects

`E(y) = $\beta_0$ + $\beta_1$(LAND) + $\beta_2$(IMP)`

- β_0 = Predicted price when LAND=0, IMP=0
- β_1 = Effect of LAND on price
- β_2 = Effect of IMP on price

Parameter Estimates

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%
Intercept	13017.945	3602.618	3.61	0.0003*	5950.1598	20085.731
LAND	1.5502941	0.024979	62.06	<.0001*	1.5012889	1.5992993
IMP	1.1130655	0.018842	59.08	<.0001*	1.0761014	1.1500297

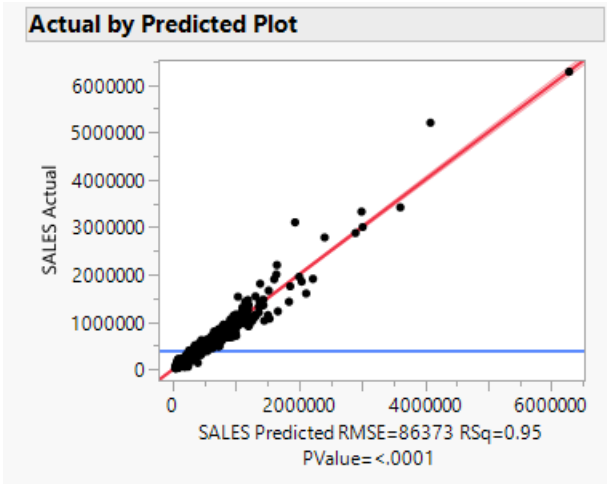
Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	2	1.9148e+14	9.574e+13	12833.43
Error	1261	9.4074e+12	7.4603e+9	Prob > F
C. Total	1263	2.0089e+14		<.0001*

Summary of Fit

RSquare	0.953171
RSquare Adj	0.953097
Root Mean Square Error	86372.98
Mean of Response	380277.5
Observations (or Sum Wgts)	1264

R2 = 0.9532, R2Adj = 0.9531, MSE = 7.4603E+9, S= 86372.98. F-test = 12833.43 ,P-value < 0.0001



Model 2: Main effects with neighborhood differences

This model includes NBHD (neighborhood) as a categorical variable.

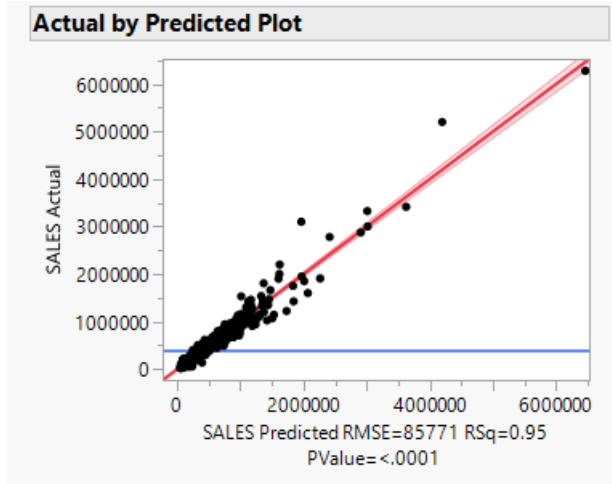
$$E(y) = \beta_0 + \beta_1(LAND) + \beta_2(IMP) + \beta_3 N_1 + \ldots + \beta_{13} N_{11}$$

- β_0 = Predicted price when LAND=0, IMP=0, and in the base neighborhood
- β_1 = Effect of LAND on price (same for all neighborhoods)
- β_2 = Effect of IMP on price (same for all neighborhoods)
- β_3 = Difference between ARBORGRN and base neighborhood (when LAND=0, IMP=0)
- ...
- β_{13} = Difference between TOWN&CTY and base neighborhood (when LAND=0, IMP=0)

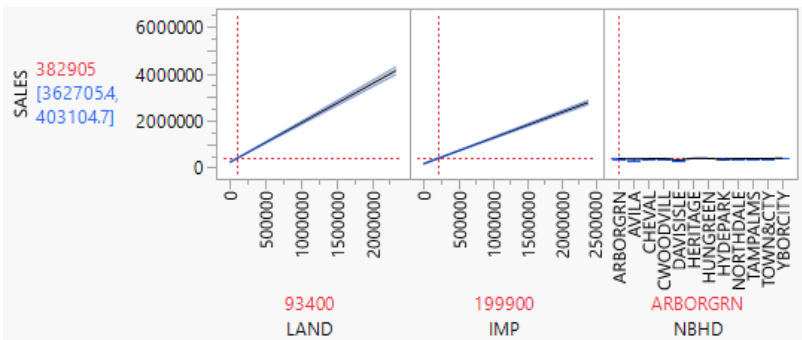
Where:

- $N_1 = 1$ if ARBORGRN, 0 otherwise

- $N_2 = 1$ if AVILA, 0 otherwise
- $N_3 = 1$ if CHEVAL, 0 otherwise
- ...
- $N_{11} = 1$ if TOWN&CTY , 0 otherwise



- Base level: The 12th neighborhood (YBORCITY)



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	13	1.9169e+14	1.475e+13	2004.392
Error	1250	9.1959e+12	7.3567e+9	Prob > F
C. Total	1263	2.0089e+14		<.0001*

Summary of Fit

RSquare	0.954224
RSquare Adj	0.953748
Root Mean Square Error	85771.12
Mean of Response	380277.5
Observations (or Sum Wgts)	1264

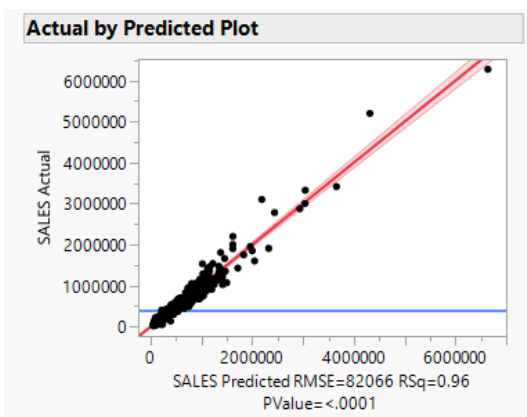
R2 = 0.9542, R2Adj = 0.9537, MSE = 7.3567e+9, F-test = 2004.392 ,P-value < 0.0001

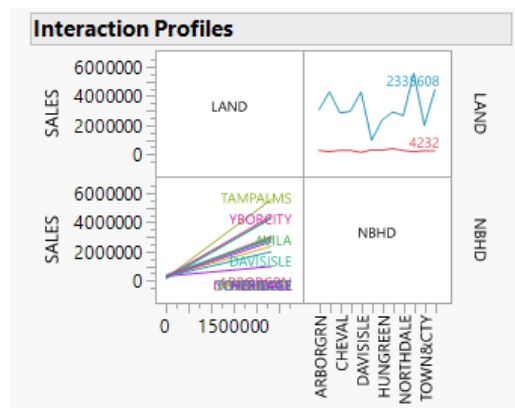
Indicator Function Parameterization

Term	Estimate	Std Error	t Ratio	Prob>|t|	Lower 95%	Upper 95%
Intercept	22997.916	6105.334	3.77	0.0002*	11020.084	34975.748
LAND	1.6723609	0.038097	43.90	<.0001*	1.5976205	1.7471013
IMP	1.0968015	0.021835	50.23	<.0001*	1.0539635	1.1396396
NBHD[ARBORGRN]	-15542.02	12143.44	-1.28	0.2008	-39365.8	8281.7507
NBHD[AVILA]	-54363.39	20747.68	-2.62	0.0089*	-95067.51	-13659.27
NBHD[CHEVAL]	-23213.99	11520.01	-2.02	0.0441*	-45814.68	-613.3011
NBHD[CWOODVILL]	-11093.45	10735.24	-1.03	0.3016	-32154.52	9967.6273
NBHD[DAVISISLE]	-78042.44	15782.86	-4.94	<.0001*	-109006.2	-47078.63
NBHD[HERITAGE]	-6513.901	9592.111	-0.68	0.4972	-25332.31	12304.513
NBHD[HUNGREEN]	-7421.149	10311.15	-0.72	0.4718	-27650.23	12807.93
NBHD[HYDEPARK]	-41090.88	17331.74	-2.37	0.0179*	-75093.4	-7088.368
NBHD[NORTHDALE]	-14220.62	11681.82	-1.22	0.2237	-37138.75	8697.5132
NBHD[TAMPALMS]	-19830.3	9746.779	-2.03	0.0421*	-38952.15	-708.4472
NBHD[TOWN&CTY]	-16108.89	9102.104	-1.77	0.0770	-33965.98	1748.1943

Model 3: Neighborhood interactions with land value

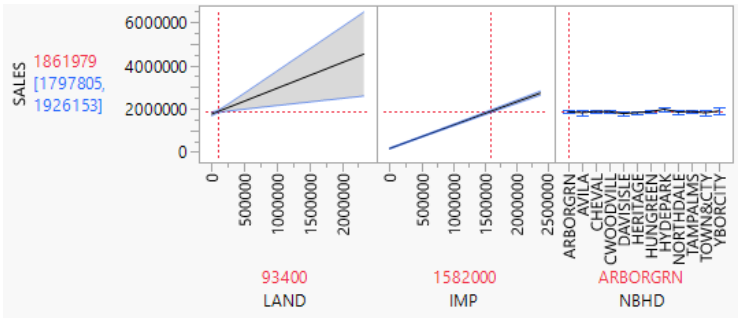
`E(y) = Model 2 + $\beta_{14}$(LANDN$_1$) + ... + $\beta_{24}$(LANDN$_{11}$)` **(Adds LANDNBHD interactions)**





Where:

- $N_1, N_2, \ldots, N_{11}$ = dummy variables for 11 neighborhoods
- **Base neighborhood** = the 12th neighborhood (no dummy variable)



Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	24	1.9255e+14	8.023e+12	1191.236
Error	1239	8.3444e+12	6.7348e+9	Prob > F
C. Total	1263	2.0089e+14		<.0001*

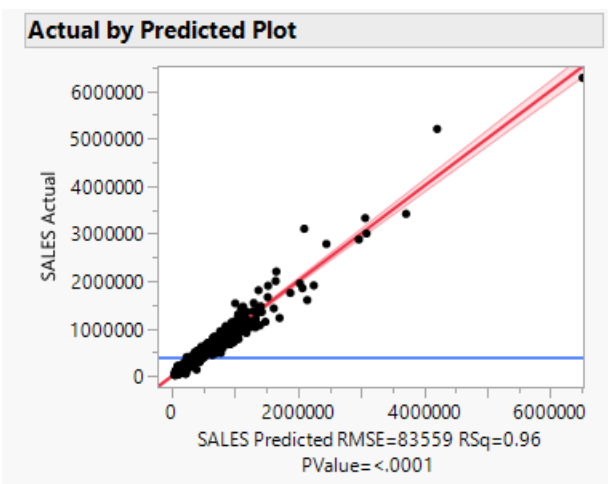
Summary of Fit

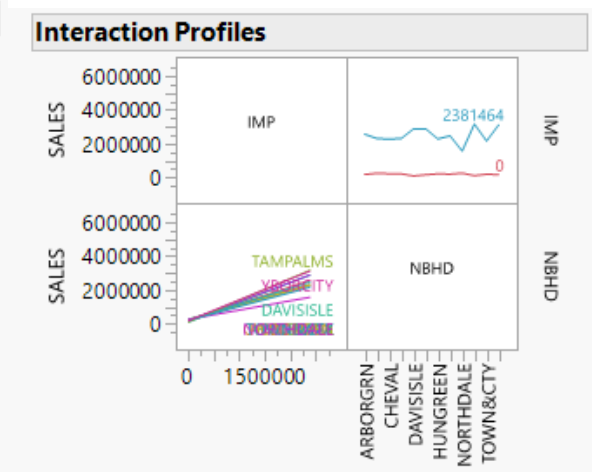
RSquare	0.958463
RSquare Adj	0.957658
Root Mean Square Error	82065.75
Mean of Response	380277.5
Observations (or Sum Wgts)	1264

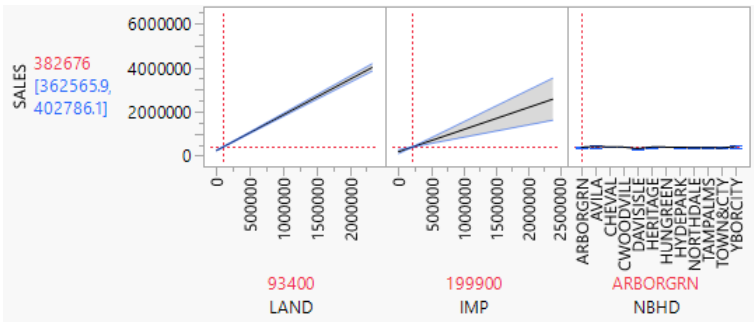
R^2 = 0.9585, R^2Adj = 0.9577, MSE = 6.7348e+9, F-test = 1191.236 ,P-value < 0.0001

Model 4: Neighborhood interactions with improvement value

E(y) = Model 2 + β_{25}(IMPN$_1$) + ... + β_{35}(IMPN$_{11}$)**(Adds IMPNBHD interactions)**







Analysis of Variance

Source	DF	Sum of Squares	Mean Square	F Ratio
Model	24	1.9224e+14	8.01e+12	1147.212
Error	1239	8.6508e+12	6.9821e+9	Prob > F
C. Total	1263	2.0089e+14		<.0001*

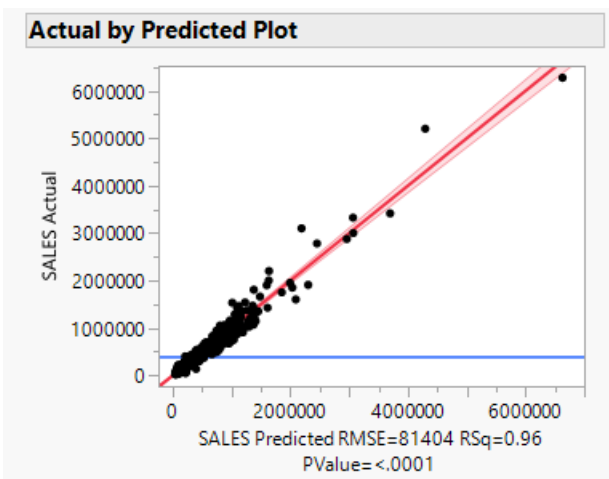
Summary of Fit

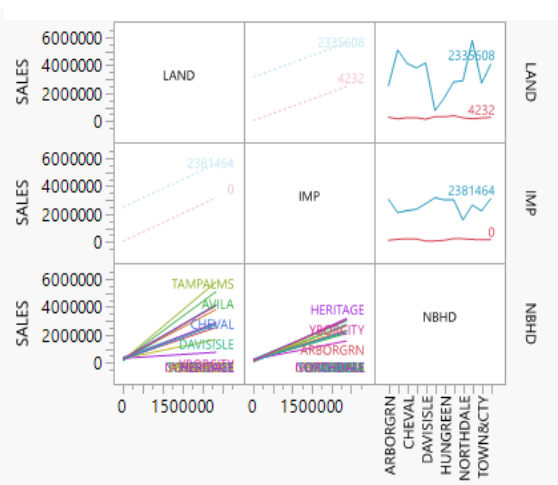
RSquare	0.956937
RSquare Adj	0.956103
Root Mean Square Error	83558.99
Mean of Response	380277.5
Observations (or Sum Wgts)	1264

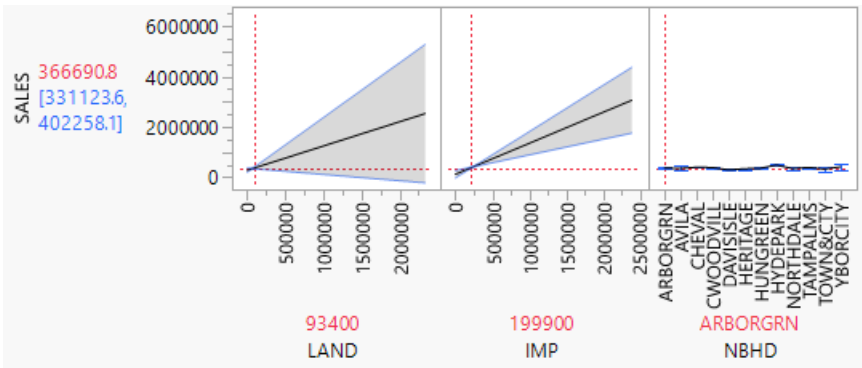
R^2 = 0.9569, R^2Adj = 0.9561, MSE = 6.9821e+9, F-test = 1147.212,P-value < 0.0001

Model 5: Full interaction model

E(y) = Model 2 + LAND*NBHD + IMP*NBHD **(Includes all interactions)**







Summary of Fit	
RSquare	0.959493
RSquare Adj	0.958339
Root Mean Square Error	81403.59
Mean of Response	380277.5
Observations (or Sum Wgts)	1264

Analysis of Variance				
Source	DF	Sum of Squares	Mean Square	F Ratio
Model	35	1.9275e+14	5.507e+12	831.0830
Error	1228	8.1374e+12	6.6265e+9	Prob > F
C. Total	1263	2.0089e+14		<.0001*

R^2= 0.9595, R^2Adj = 0.9583, MSE = 6.6265e+9, F-test = 831.08,P-value < 0.0001

Model Comparison Framework

Global F-Test: Used to assess the overall significance of each model.

Partial F-Test: Use **nested model comparison**

- **Model 2** is the **reduced model** (includes main effects + NBHD indicators).
- **Models 3–5** are **complete models** (add interaction terms**)**.
- **Tool Used**: JMP's **Custom Test**.
- Perform **Partial F-Test**:
    - Test if the added interaction terms significantly improve the model.
    - **Tested Parameters**: Interaction terms (e.g., LAND×NBHD, IMP×NBHD).
    - If **F is large** and **p < 0.05**, the full model is significantly better

- If **$p > 0.05$**, prefer the simpler model (Principle of Parsimony).

Following hierarchical model-building principles, partial F-tests were used for model selection rather than individual t-tests, as they more appropriately test sets of related parameters like all neighborhood indicators or all interaction terms simultaneously.

Goodness-of-Fit: Assessed using Adjusted R^2 (proportion of variance explained, adjusted for parameters) and Mean Square Error (MSE) (average squared prediction error).

Evaluate Model Fit compare: All tests used $\alpha = 0.05$.

Metric	**What to Look For**
Adjusted R^2	Higher is better, but beware of overfitting
RMSE / MSE	Lower values indicate better fit
CV ($CV=\bar{y}s\times100$)	Should be <10% ideally
Significance of interactions	Use T-tests and plots
Interpretability	Can you explain the model clearly?

3. Results

Initial Model Fit and Comparison

Table 1 *:* Model Summary and Goodness-of-Fit

Model	Description	Adj R^2	RMSE	CV (%)	Global F	p-value
1	LAND + IMP	0.9531	86,373	22.71	12833.43	<.001
2	+ Neighborhood	0.9537	85,771	22.55	2004.39	<.001
3	+ LAND×NBHD	0.9577	82,066	21.58	1191.24	<.001
4	+ IMP×NBHD	0.9561	83,559	21.97	1147.21	<.001
5	Full Interaction	0.9583	81,404	21.41	831.08	<.001

Initial Model Fit and Comparison

Goodness-of-fit statistics for all five models are presented in Table 1. As expected, model complexity increases from Model 1 to Model 5, and the standard descriptive metrics reflect this:

- Adjusted R^2 increases with model complexity, with Model 5 (the full interaction model) achieving the highest value (0.9583).
- RMSE and CV generally decrease with added complexity, with Model 5 again showing the best (lowest) values.

While these metrics **suggest Model 5 may have the best fit**, they do not account for statistical significance or the principle of parsimony. The improved fit from adding interactions in Models 3, 4, and 5 is relatively small compared to the substantial increase in complexity. **To determine if this improvement is statistically significant, we proceed to formal nested model comparisons using partial F-tests.**

Nested Model Comparisons

To determine whether adding interaction terms significantly improves the model's predictive ability, we performed Partial F-Tests using JMP's Custom Test feature**. (Please see Figure 5-8)**. This method calculates **the F-ratio and p-value** for nested model comparisons, where the reduced model is a subset of the full model.

Table 2: Partial F-Test Results for Nested Model Comparisons

Comparison	Hypothesis	F-ratio	p-value	Conclusion
Model 2 vs Model 1	NBHD indicators	11.4993	0.0007	**Significant** - Neighborhood effects matter
Model 3 vs Model 2	LAND×NBHD interactions	11.4933	1.2416e-20	**Highly Significant** - Land effects vary by NBHD
Model 4 vs Model 2	IMP×NBHD interactions	7.0964	8.81e-12	**Highly Significant** - Improvement effects vary by NBHD
Model 5 vs Model 3	LAND×NBHD IMP×NBHD interactions	7.2605	2.64e-21	**Highly Significant** - improvement over Model 3

Key Findings from Nested Model Comparisons

Neighborhood baseline effects are significant

- Model 2 vs Model 1 is significant (**p = 0.0007**)
- This confirms that different neighborhoods have systematically different baseline prices, even after controlling for land and improvement values.

Land valuation varies significantly by neighborhood

- Model 3 vs Model 2 is highly significant (**p = 1.24e-20**)
- The relationship between land value and sale price differs substantially across neighborhoods. Each dollar of appraised land contributes differently to market value depending on location.

Improvement valuation also varies by neighborhood

- Model 4 vs Model 2 is highly significant ($p = 8.81e-12$)
- The return on home improvements differs significantly across neighborhoods, indicating that improvements are not valued uniformly throughout Tampa.

Full interaction model provides significant improvement

- Model 5 vs Model 3 is highly significant ($p = 2.64e-21$)
- Allowing both land and improvement effects to vary by neighborhood significantly improves model fit beyond just land interactions alone.

Summary

The partial F-tests reveal that both land and improvement valuation practices vary significantly across neighborhoods. This contradicts the hypothesis of consistent appraisal practices and **suggests that appraisers apply different valuation criteria in different areas.**

Final Model Selection

Based on the hierarchical model comparison and statistical significance, **Model 5** (SALES ~ LAND + IMP + NBHD + LAND×NBHD + IMP×NBHD) is selected as the final and optimal predictive model. This decision is strongly supported by the Partial F-Test results:

- Model 5 has the highest Adjusted R^2 (0.9583) and lowest RMSE (81,403.59)
- All partial F-tests for interaction terms are highly statistically significant ($p < 0.0001$)

- The full interaction model provides significantly better explanatory power than simpler models
- While more complex, Model 5 captures the reality that both land and improvement valuation practices vary systematically across neighborhoods

Although the improvement in R^2 appears modest, the highly significant partial F-tests ($p < 0.0001$) indicate these interaction effects represent real, meaningful variations in appraisal practices

Model 5 offers the most accurate representation of Tampa's real estate market, revealing that location affects not just baseline price levels but also how physical attributes are converted into market value.

Key parameter estimates for the final selected model (Model 5) show substantial variation in coefficients across neighborhoods, supporting the conclusion that valuation practices differ by location.

4. Conclusion

This study reveals significant variations in appraisal practices across Tampa neighborhoods. While neighborhood location clearly influences baseline housing prices, the way land and improvement values translate into market prices also varies substantially across neighborhoods. In other words, location affects both levels and slopes, providing evidence of neighborhood-specific valuation practices rather than consistent appraisal criteria.

Three key insights emerge:

First, appraisal practices vary systematically by neighborhood. The analysis found strong evidence that both land and improvement valuation differ across locations. Each additional dollar of appraised land contributes differently to market value depending on the neighborhood, as do improvements.

Second, neighborhood affects both baseline prices and valuation rates. Location influences not just the starting price point but also how physical attributes translate into market value, suggesting complex neighborhood-specific factors at play.

Third, the full interaction model is statistically justified. While more complex, Model 5 significantly outperforms simpler models, indicating that the real-world relationship between appraised values and sale prices is neighborhood dependent.

Implications

- For Appraisers & Tax Authorities:

  The results suggest potential inconsistencies in appraisal practices across neighborhoods. Further investigation is needed to determine whether these variations reflect legitimate market differences or systematic assessment biases that could affect tax equity.

- For Buyers, Sellers, and Investors:

Both location and the neighborhood-specific valuation of physical attributes matter significantly. Investment strategies should account for the fact that the same physical improvements may yield different returns depending on location, and systematic undervaluation or overvaluation may exist in specific areas.

Limitations and Future Research

While the model explains over 95% of the variation in sale prices, it is important to note that unobserved neighborhood characteristics—such as school quality, crime rates, or specific amenities—are likely captured by the neighborhood fixed effects and interactions. Future research could incorporate these additional variables directly to parse out their specific influence on property values.

Overall, this analysis suggests a complex appraisal landscape in which property characteristics and location interact in ways that vary systematically across neighborhoods. The market appears to reward improvements and land value differently depending on location, and neighborhood differences reflect both baseline premiums and varying valuation criteria.

Figure 5: Partial F-test: Model 2 vs Model 1

Custom Test

Parameter	
Intercept	0
LAND	0
IMP	0
NBHD[ARBORGRN]	1
NBHD[AVILA]	1
NBHD[CHEVAL]	1
NBHD[CWOODVILL]	1
NBHD[DAVISISLE]	1
NBHD[HERITAGE]	1
NBHD[HUNGREEN]	1
NBHD[HYDEPARK]	1
NBHD[NORTHDALE]	1
NBHD[TAMPALMS]	1
NBHD[TOWN&CTY]	1
=	0

Value	-23953.41908
Std Error	7063.6844127
t Ratio	-3.391065864
Prob>|t|	0.0007179544
SS	84596935312

Sum of Squares	84596935312
Numerator DF	1
F Ratio	11.499327691
Prob > F	0.0007179544

Figure 6: Partial F-test: Model 3vs Model 2

Response SALES

Prediction Profiler

Custom Test

Parameter											
Intercept	0	0	0	0	0	0	0	0	0	0	0
LAND	0	0	0	0	0	0	0	0	0	0	0
IMP	0	0	0	0	0	0	0	0	0	0	0
NBHD[ARBORGRN]	0	0	0	0	0	0	0	0	0	0	0
NBHD[AVILA]	0	0	0	0	0	0	0	0	0	0	0
NBHD[CHEVAL]	0	0	0	0	0	0	0	0	0	0	0
NBHD[CWOODVILL]	0	0	0	0	0	0	0	0	0	0	0
NBHD[DAVISISLE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HERITAGE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HUNGREEN]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HYDEPARK]	0	0	0	0	0	0	0	0	0	0	0
NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	0	0
NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[ARBORGRN]	1	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[AVILA]	0	1	0	0	0	0	0	0	0	0	0
LAND*NBHD[CHEVAL]	0	0	1	0	0	0	0	0	0	0	0
LAND*NBHD[CWOODVILL]	0	0	0	1	0	0	0	0	0	0	0
LAND*NBHD[DAVISISLE]	0	0	0	0	1	0	0	0	0	0	0
LAND*NBHD[HERITAGE]	0	0	0	0	0	1	0	0	0	0	0
LAND*NBHD[HUNGREEN]	0	0	0	0	0	0	1	0	0	0	0
LAND*NBHD[HYDEPARK]	0	0	0	0	0	0	0	1	0	0	0
LAND*NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	1	0	0
LAND*NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	1	0
LAND*NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	1
=	0	0	0	0	0	0	0	0	0	0	0
Value	-0.069005538	0.5029440505	-0.161862787	-0.105550413	0.5236311678	-0.974913274	-0.371345337	-0.183049111	-0.228183944	1.0524754284	-0.518873897
Std Error	0.4210839708	0.2786578	0.1936975797	0.3559124852	0.1342445335	0.3452856341	0.2323191142	0.1575670539	0.7510922915	0.1809963342	0.6805425436
t Ratio	-0.163875955	1.8048805757	-0.835646928	-0.296562828	3.9005772096	-2.823497933	-1.598427826	-1.16172199	-0.303802804	5.8148991411	-0.762441529
Prob>|t|	0.8698555239	0.0713360769	0.4035146753	0.7668499972	0.0001011161	0.0048263128	0.1102027641	0.2455721534	0.7613291398	7.7118088e-9	0.44594165
SS	180864931.91	21939202735	4702941113.3	592321271.23	102466442217	53690672977	17207190301	9089255657.4	621594919.17	227723721203	3915047053.2

Sum of Squares	851454907278
Numerator DF	11
F Ratio	11.493308615
Prob > F	1.241597e-20

Figure 7: Partial F-test: Model 4 vs Model 2

Response SALES

Prediction Profiler

Custom Test

Parameter											
Intercept	0	0	0	0	0	0	0	0	0	0	0
LAND	0	0	0	0	0	0	0	0	0	0	0
IMP	0	0	0	0	0	0	0	0	0	0	0
NBHD[ARBORGRN]	0	0	0	0	0	0	0	0	0	0	0
NBHD[AVILA]	0	0	0	0	0	0	0	0	0	0	0
NBHD[CHEVAL]	0	0	0	0	0	0	0	0	0	0	0
NBHD[CWOODVILL]	0	0	0	0	0	0	0	0	0	0	0
NBHD[DAVISISLE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HERITAGE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HUNGREEN]	0	0	0	0	0	0	0	0	0	0	0
NBHD[HYDEPARK]	0	0	0	0	0	0	0	0	0	0	0
NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	0	0	0
NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	0	0
NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	0
IMP*NBHD[ARBORGRN]	1	0	0	0	0	0	0	0	0	0	0
IMP*NBHD[AVILA]	0	1	0	0	0	0	0	0	0	0	0
IMP*NBHD[CHEVAL]	0	0	1	0	0	0	0	0	0	0	0
IMP*NBHD[CWOODVILL]	0	0	0	1	0	0	0	0	0	0	0
IMP*NBHD[DAVISISLE]	0	0	0	0	1	0	0	0	0	0	0
IMP*NBHD[HERITAGE]	0	0	0	0	0	1	0	0	0	0	0
IMP*NBHD[HUNGREEN]	0	0	0	0	0	0	1	0	0	0	0
IMP*NBHD[HYDEPARK]	0	0	0	0	0	0	0	1	0	0	0
IMP*NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	1	0	0
IMP*NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	1	0
IMP*NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	1
=	0	0	0	0	0	0	0	0	0	0	0
Value	0.0312054898	-0.107462014	-0.107996683	-0.091435067	0.2038734587	0.1727278727	-0.102385411	-0.017796036	-0.418593688	0.3110681869	-0.143912285
Std Error	0.2092150925	0.0877026687	0.0625460625	0.112221944	0.0462310664	0.1660267448	0.0873192152	0.0889124237	0.2315969817	0.0612757561	0.1614469961
t Ratio	0.1491550605	-1.225299251	-1.726674373	-0.814769941	4.4098800775	1.0403617377	-1.172541581	-0.200152413	-1.807422897	5.0765295522	-0.891390291
Prob>|t|	0.8814555529	0.2206953621	0.0844754188	0.4153605874	0.0000112397	0.2983748392	0.2412050778	0.8413942292	0.0709388224	4.4318186e-7	0.3728928931
SS	155332511.02	10482641029	20816478580	4635070826.1	135781291845	7557099150.2	9599373348.9	279710028.25	22808983811	179936892060	5547817629.6

Sum of Squares	545028528671
Numerator DF	11
F Ratio	7.0964340766
Prob > F	8.813399e-12

Figure 8: Partial F-test: Model 5 vs Model 3

Response SALES

Custom Test

Parameter																		
NBHD[HUNGREEN]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NBHD[HYDEPARK]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[ARBORGRN]	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[AVILA]	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[CHEVAL]	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[CWOODVILL]	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[DAVISISLE]	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[HERITAGE]	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[HUNGREEN]	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[HYDEPARK]	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0	0
LAND*NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0	0
LAND*NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0	0
LAND*NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0	0
IMP*NBHD[ARBORGRN]	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0	0
IMP*NBHD[AVILA]	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0	0
IMP*NBHD[CHEVAL]	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0	0
IMP*NBHD[CWOODVILL]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0	0
IMP*NBHD[DAVISISLE]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0	0
IMP*NBHD[HERITAGE]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1	0
IMP*NBHD[HUNGREEN]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1
IMP*NBHD[HYDEPARK]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IMP*NBHD[NORTHDALE]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IMP*NBHD[TAMPALMS]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
IMP*NBHD[TOWN&CTY]	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
=	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Value	-0.375696295	0.7776604365	0.3332245122	0.1887574248	0.3891953294	-1.157033946	-0.756503	-0.305536199	-0.202250827	1.0733750296	-0.267662218	0.2107902022	-0.223261882	-0.177358389	-0.128186799	0.0979253478	0.2760620977
Std Error	0.5841513912	0.3035155946	0.2832413086	0.4538831767	0.1517146173	0.3557808278	0.3939503654	0.1729132517	0.7490302265	0.2843911558	0.7293965168	0.2885667459	0.0950622166	0.0926041756	0.1415116123	0.0544251399	0.1676471968
t Ratio	-0.64314885	2.5621762124	1.1764686228	0.4158722651	2.5653120062	-3.252097514	-1.920300288	-1.766991231	-0.270016909	3.7742911752	-0.366963938	0.73047295	-2.348586958	-1.915231009	-0.905839436	1.7992668087	1.6466848414
Prob>|t|	0.5202475934	0.01052031	0.2396356658	0.6775761386	0.010426211	0.0011767036	0.0550513787	0.0774780545	0.7871926138	0.0001681236	0.7137091412	0.4652404759	0.0190026657	0.0556951235	0.3651985385	0.0722219118	0.0998787043
SS	2741006759.7	43501587057	9171657098.1	1146059135.8	43608133578	70083249567	24435734958	20689781307	483135594.46	94396939741	892347246.56	3535862665.6	36551095803	24306892542	54373784248	21452516711	17968345391

Sum of Squares	1.03867e+12
Numerator DF	21
F Ratio	7.4639897783
Prob > F	3.175345e-21